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324551(24)

B. E. (Fifth Semester) EXAMINATION, April-May, 2015 (New Scheme)

(Branch : Elect.)

ELECTRICAL MACHINES—II

Time : Three Hours] [*Maximum Marks : 80*

[*Minimum Pass Marks : 28*

Note : Attempt part (a) and any *two* from parts (b), (c) and (d) of each question. Part (a) is of 2 marks and part (b), (c) and (d) of 7 marks each.

1. (a) State two electromagnetic phenomena responsible for electromagnetic energy conversion.

   (b) Explain the production of torque in synchronous and induction machines.

   (c) Show that fundamental space wave of a distributed single phase winding is given by :

$$F_{a_1} = \frac{4}{\pi} K_p K_d \frac{NPh}{P} I_a \cos\theta$$

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where I_a is instantaneons current flowing through coil of N_{Ph} number of twins in series per phase and P is number of poles.

   (d) A three phase 4 pole star connected alternator has 60 slots with 2 conductors per slot. The pitch of coil is 3 slots less than pole pitch. The flux per pole is 0.125 Wb. Calculate no load terminal voltage. The speed is 1500 r. p. m.

2. (a) Draw the synchronous impedance verses field current variation curve of an alternator.

   (b) Explain short circuit ratio. Show that it is reciprocal of synchronous reactance on per unit.

   (c) Why does regulation calculated using Ampere turn method give lower value than actual ? The no load excitation of an alternator to give rated voltage on open circuit is 100 A. In a short circuit test with full load current flowing in armature, the field current was 80 A. Calculate field current required to give rated voltage at rated current 0.8 p. f. lag.

   (d) Derive the power angle equation of an alternator (non salient pole) for input and output power. Hence comment on maximum power conditions.

3. (a) Why the alternators run by water turbines salient pole machines ?

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(b) Draw neat phasor diagram of salient pole synchronous machine working as (i) Generator (ii) Motor on lagging and leading power factor.

(c) Explain step test to measure X_d and X_q of an alternator.

(d) The synchronous reactance per phase of a three phase star connected 6600 V synchronous motor is 10 Ω with neglible resistance. For a certain load the input is 900 kW and induced e.m.f. E is 8200 V (line). Find line current and power factor.

4. (a) The no load current of an induction motor is more than that of a transformer. Explain.

(b) Show that in an induction motor

rotor input : power developed : rotor copper loss :: 1 : (1 − S) : S

where S is slip.

(c) A 400 V, 3-phase star connected induction motor gave the following test results :

No load test 400 V, 8.5 A, 1100 W

Blocked rotor test 180 V, 45 A, 520 W.

Determine the ohmic values of components in circuit model and calculate line current and p.f when slip is 5%. The stator resistance per phase is 0.5 Ω and $X_1 = X_2$.

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(d) An 8 pole 3 ϕ 50 Hz induction motor running at a slip of 4% is taking 20 kW. Stator losses amount to 0.5 kW. If the mechanical torque lost in friction is 16.25 Nm find :

(i) BHP

(ii) Efficiency

5. (a) Why are starters required while starting induction motors ?

(b) Explain cogging and crawling of induction motors.

(c) What is V/f control of speed of three phase induction motor ? In this method show that :

(i) Ship at maximum torque decreases with f increasing

(ii) The maximum torque remains constant

(d) Determine a suitable autotransformer ratio for starting a three phase induction motor with line current not exceeding twice the full load current. The short circuit current is 4 times the full load current and full load slip is 2.5%. Estimate the starting torque in terms of full load torque.

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